

ACIDS, ALKALIS, AND SALTS

In previous classes, we have learnt about the uses of different compounds in our daily life. We have also studied that a new compound is formed when two or more substances react together and change chemically. This means that compounds may have different physical and chemical properties. In this chapter, we will study the nature of compounds in the light of their behavior as acids, alkalis and salts. Do you know what stomach secretes from the lining of its wall to help in digestion? Have you ever suffered with the acidity in stomach? How did you cure it? Why some foods are sour whereas others are bitter in taste? Are these substances chemically same or different?

In this Chapter you will learn about:

- Introduction to Acids, Alkalis and Salts
- Properties of Acids, Alkalis and Salts
- Uses of Acids, Alkalis and Salts
- pH and its Range (1-14) in Aqueous Medium
- Indicators and their Uses (Natural Indicators from Fruits and Vegetables)

All the students will be able to:

- Define the terms acid, alkali and salt.
- Describe the properties of acids, alkalis and salts.
- Explain the uses of acids, alkalis and salts in daily life.
- Define indicators.
- Use indicators to identify acids, alkalis and neutral substances.
- Investigate the colour changes in the extracts of various flowers and vegetables by adding acids and alkalis.



Fig.6.1: Colours in different solutions



Fig.6.2: Acids and Alkalis around us

Exploration: Farmers are suggested to get their soils tested before selecting and putting fertilizers. Why?

INTRODUCTION TO ACIDS, ALKALIS AND SALTS

➤ Define the terms; Acids, Alkalis and Salts

Do you know why vinegar and lemon juice are sour in taste. Nobody knew this until a few hundred years ago when it was discovered that these things are acids. The term acid, in fact, comes from the Latin term *acere*, which means “sour”. Initially acids and bases are defined as substances that change some properties of water. The concept evolved in 19th century when these substances were categorized based on their tastes. For example, acids are sour-tasting stuff.

Exploration:

Have you eaten salads with vinegar? What is its taste? What do you think vinegar is?

With the passage of time, people found out that the sour-tasting stuff had some other properties in common, apart from just tasting sour; for example, it changes the colour of litmus paper and corrodes some metals. Lavoisier, the father of modern chemistry, gave a raw idea about the chemistry of acids that there is some element, or essence in the acid that would be responsible for its acidity. In 1840s it was proposed that acidity was generated by hydrogen, because it is the component all acids have in common. Swedish chemist, August Arrhenius in 1903 defined acids and bases on their properties of ionization in the solution and received Nobel prize on his discovery. According to Arrhenius;

Acids are those substances which produce hydrogen ion (H^+) in the aqueous solution.

For example, hydrochloric acid (HCl) is a common acid when dissolved in water, dissociates into H^+ and Cl^- . H^+ ion combines with water to form hydronium ion (H_3O^+)



Whereas

Bases are those substances which produce hydroxyl ion (OH⁻) in the aqueous solution.

For example, sodium hydroxide (NaOH) is a common base which dissociates into Na⁺ and OH⁻ when dissolved in water.



DO YOU KNOW?

All alkalis are bases but all bases are not alkalis

This is important to note that those bases which are soluble in water are termed as alkali. All acids have H⁺ ion and bases have OH⁻ ions.

Activity 6.1:

Identification of Acidic and Basic Substances

Instructions: Identify which of the following formulae represent acidic or basic substance. Mark the checks in the correct column. Also, justify your response during discussion with your classmate sitting next to you.

Substance (Formulae)	Acids	Bases	Reasons
H ₂ SO ₄			
KOH			
HCl			
CH ₃ COOH			
NH ₄ OH			
Ca(OH) ₂			

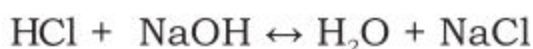
Salt

When acid is mixed with base, at certain point acidic nature of acid and basic nature of base are destroyed. The resulting solution is neither acidic nor basic but neutral. This process is called *neutralization* and that point is called *neutralization point*. If you touch the

DO YOU KNOW?

H₂O is the main product of neutralization as it is formed by the H⁺ from acid and OH⁻ from base.

test tube immediately after neutralization point, you will feel test tube hot as heat always released during neutralization reaction and hence increase the temperature of reacting mixture. A new substance formed as a result of neutralization is called “salt” along with H_2O .



Activity 6.2:

Match with the Correct Option.

Instructions: Identify which of the acid-base pair mentioned in column B will neutralize to form the following salts in column A. Write your answers in the space given below.

	Salts		Acid & Base
1	Sodium Benzoate ($\text{C}_6\text{H}_5\text{COONa}$)	a	Potassium hydroxide (KOH) and Hydrochloric Acid (HCl)
2	Potassium Chloride (KCl)	b	Sodium hydroxide (NaOH) and Nitric Acid (HNO_3)
3	Sodium Acetate (CH_3COONa)	c	Sodium hydroxide (NaOH) and Acetic Acid (CH_3COOH)
4	Sodium Chloride (NaCl)	d	Sodium hydroxide (NaOH) and Benzoic Acid ($\text{C}_6\text{H}_5\text{COOH}$)
5	Sodium Sulphate (Na_2SO_4)	e	Sodium hydroxide (NaOH) and Hydrochloric Acid (HCl)

Answers: 1. ____ 2. ____ 3. ____ 4. ____ 5. ____

PROPERTIES OF ACIDS, ALKALIS AND SALTS

➤ Describe the properties of acids, alkalis and salts.

Acids are sour in taste as you have studied that vinegar and lemon are sour in taste. Whereas Bases are bitter.

There are many acids and bases which occur in nature. Several acids and bases are also prepared. Few examples are mentioned below,

Formed Naturally		Prepared in Laboratory	
Name of Acid	Found in	Name of Base	Found in
Tartaric Acid	Tamarind, grapes	Calcium hydroxide	Lime water
Citric Acids	Citrus fruits like lemons and oranges	Sodium hydroxide / potassium hydroxide	Soap
Lactice Acid	Yogurt	Ammonium hydroxide	Window cleaner
Ascorbic acid	Amla, Citrus Fruits	Magnesium	Milk of Magnesia

Similarly, there are naturally occurring salts which are processed from salt mines or by the evaporation of sea water. Several soluble and insoluble salts can be prepared in the laboratory through neutralization process.

Activity 6.3:

What is What

Instructions: Using the above information, taste the following food items, write whether these are sour, bitter or of any other taste and predict their acidic or basic nature.

Precautions: Don't taste things unless it is asked to do so. Every acidic and alkaline substances around us are not safe to taste and touch.

Food Item	Sour / Bitter / Saltish	Acidic / Basic / Salt
Yogurt		
Common salt		
Lemon juice		
Baking Soda		
Tamarind (Imli)		

Electric Conductivity: We all know that water is a bad conductor of electricity. But the tap water contains charged particles like magnesium and calcium ions due to the presence of salts into it. Hence it is dangerous to use electrical appliances if the surface area is wet.

Activity 6.4:

Electrical Conductivity

Note for Teachers: Arrange the materials at local site.

Materials required: Two nails, cork, beakers, 6 volt battery (electricity source), bulb, switch, 10 ml of each of the following:

1.0 M HCl, 1.0 M NaOH, distilled water, tap water, NaCl solution, vinegar, sugar and water solution, NaCl (solid)

Procedure:

1. Fix two nails on the cork and place the cork in the beaker as shown in the figure 6.3.
2. Connect the nails to the two terminals of 6V battery through a bulb and switch.
3. Pour some HCl in the beaker and switch on the current.
4. Record your observation in the following grid.
5. Repeat the experiment separately with other substances.
6. What do you observe? Does the bulb glow in all cases?

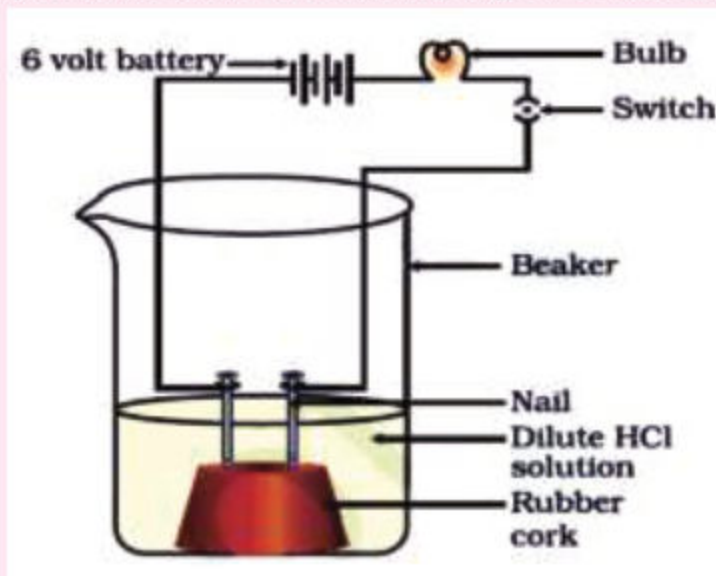


Fig 6.3

Observation:

Substances	Intensity of Light Bulb (very bright, bright, dim)	Conducts Electricity? (yes / no)
HCl		
NaOH		
Distilled water		
Tap water		
Vinegar		
NaCl solid		

Based on your above observations this is now clear that acids and alkalis are good conductor of electricity. Similarly salts also conduct electricity in molten and solution form but not in a solid state.

Exploration

Can you discuss in your group?

- Why did the bulb glow in the solution of HCl and NaOH but not in sugar and water solution?
- Why did the bulb glow brightly in the HCl than the vinegar?
- Why did the bulb glow in the solution of NaCl and not in the solid form?

Litmus Test: If the blue litmus paper turns into red or red litmus remain unaffected, then the substance is acidic in nature. Whereas if red litmus paper turns into blue or blue litmus remains unaffected the substance is basic in nature. If neither the

DO YOU KNOW?

Predictions and observations are two important scientific skills

red litmus paper nor the blue paper changes its colour then the substance is neutral.

Activity 6.5:

Acid or Base

Material required: Test tubes, NaCl solution, blue and red litmus paper strips, shampoo, vinegar, lemon juice, tooth paste

Procedure:

1. Take three liquids in the three different test tubes.
2. Before experiment, predict what would happen if the blue and red litmus papers are dipped in the given liquids placed separately in three test tubes.
3. Register your prediction in the column below.
4. Now perform the experiment and dip blue and red litmus paper strips in the three test tubes containing different liquids.

Liquids	Predictions	Observations
NaCl		
Shampoo		
Vinegar		
Tooth paste		
Lemon juice		

USES OF ACIDS, ALKALIS AND SALTS

➤ Explain the uses of acids, alkalis and salts in daily life.

Do you know that your stomach produces hydrochloric acid to help in digestion of proteins, kills harmful bacteria and minimizes the risk of food poisoning. Certain digestive enzymes require low pH less than 2 to function properly. On the other hand pancreatic fluids are alkaline, as

enzymes in intestine requires alkaline pH for breaking down starches, fats and proteins. Further, the uses of acids and alkalis are very common in industries and homes.

Following table shows uses of some common acids and alkalis

Acid		Base	
Example	Uses	Example	Uses
Benzoic acid	Its salt is used to preserve food	Ammonia	Production of fertilizers, used in the manufacture of nitric acid
Carbonic acid	To make carbonated drinks	Aluminum hydroxide	Manufacture other aluminum compound and to make gastric medicine (antacid)
Acetic Acid	Main compound of vinegar	Calcium hydroxide	To make cement, limewater, neutralize the acidity of soil and application of sewage treatment.
Hydrochloric acid	Household cleaning, leather processing	Sodium hydroxide	Used in the manufacturing of soaps, detergents, and cleaners.
Nitric acid	Production of fertilizers, explosives, purification and extraction of gold	Magnesium hydroxide	Suspension of magnesium hydroxide in water is used as an antacid

Salt is essential for life in general, and saltiness is one of the basic human tastes. The animal tissues contain larger quantities of salt than plant tissues. Salts can be used in water conditioning, food, agriculture and preparation of industrial chemicals ranging from plastic, paper, glass, polyester, rubber and fertilizers to household bleach, soaps, detergents and dyes.

Exploration

The sting of an ant contains formic acid? When an ant bite, it injects the acidic liquid into the skin. The effect of the sting can be neutralized by rubbing basic salts for example moist baking soda or calamine solution which contains zinc carbonate. Are all salts neutral or basic in nature?

pH AND ITS RANGE IN AQUEOUS MEDIUM

- Explain pH range for acids, and bases.

The **pH scale** is a measure of the acidity or alkalinity of a solution. The pH of a solution, determines how acidic or alkaline it is by reference to the pH scale (Fig 6.4) or whether the solution is neutral. pH scale ranges between 1-14. The smaller the pH number, the more acidic it is, the greater the pH number, the more alkaline it is, and if the pH is close to 7, the solution is more or less neutral.

Note: Check the following substances with litmus paper to get an idea which of the following substances are acidic or alkaline.

1	2	3	4	5	6	7	8	9	10	11	12	13	14
Hydrochloric acid	Lemon juice	Vinegar	Soda	Rain water	Milk	Pure water	Egg white	Baking Soda	Milk of Magnesia	Ammonia	Mineral lime	Bleach	Sodium hydroxide

Fig 6.4:

INDICATORS AND THEIR USES

- Define indicators.
- Use indicators to identify acids, alkalis and neutral substances.
- Investigate the colour changes in the extracts of various flowers and vegetables by adding acids and alkalis.

An indicator is a dye substance or mixture of colored substances that when added to the solution gives a different colour depending on the pH of the solution. This helps to indicate whether the solution is acidic or alkaline. There are different chemicals or homemade indicators which are used to determine the pH value of any solution. The most accurate way of measuring pH is by using electronic pH meter.

Common Chemical Indicators

Some common chemically prepared indicators are Phenolphthalein, Universal indicator, methyl orange (Fig 6.5 & 6.6)

Indicators	Colour in Acidic medium	Colour in Basic medium
Phenolphthalein	Colourless	Pink
Bromophenol blue	Yellow	Blue
Methyl orange	Orange	Yellow



Fig 6.5: Methyl Orange



Fig 6.6: Phenolphthalein

Activity 6.6:

Using Indicator(s) for Identification

Material used: HCl, NaOH, phenolphthalein, methyl orange, beet juice, beaker, glass dishes

Procedure:

1. Take HCl, NaOH and NaCl (aqueous solution) in three beakers. (don't label them as acid, base or neutral)
2. Mark the beakers as solution 1, 2 and 3.
3. Take small quantity of each into different glass dishes separately and use the above indicators one by one in order to observe the colour change.
4. Record your observations in the given table.
5. Can you predict which solution is HCl (acid), NaOH (base) or NaCl (neutral).

Indicator(s)	Colour in solution 1	Colour in solution 2	Colour in solution 3
Phenolphthalein			
Methyl Orange			
Beet Juice			

SUMMARY

- Acidic substances generate H^+ ions in the medium, while basic ones liberate OH^- ions in the medium.
- Strength of an acid depends on the concentration of hydronium ions present in a solution. Greater the numbers of hydronium ions present, greater is the strength of the acid.
- Strength of a base depends on the concentration of hydroxyl ions.
- Those bases which dissolve in water are called alkali.
- Salts are important compounds that are obtained by treating an acid with a base. Important salts used in everyday life and industrial applications are Sodium chloride ($NaCl$), Sodium carbonate (Na_2CO_3), Sodium Bicarbonate ($NaHCO_3$).
- Acidic and basic solutions are good conductors of electricity.
- For acid, blue litmus paper turns into red whereas for bases red litmus paper turns into blue. Both litmus paper will remain unchanged in case if the solution is neutral.
- The strength of an acid or base is expressed on a 14 point scale (ranges from 1 to 14) known as pH Scale.
- An acidic solution has a pH less than 7 and a basic solution a pH more than 7, while a neutral solution has a pH of exactly 7.

EXERCISE

The table shows the pH of four substances. Answer the questions below while considering the following table.

Substance	pH
Blood	7.3
Orange juice	4
Ammonia	11
Milk	6.5

- i) Which best describes milk?
- Slightly basic
 - Strongly basic
 - Slightly acidic
 - Strongly acidic
- ii) Which lists the substances from least to most acidic?
- Ammonia, blood, milk, orange juice
 - Orange juice, milk, blood, ammonia
 - Ammonia, milk, blood, orange juice
 - Orange juice, blood, milk, ammonia
- iii) Which substance is the most alkaline?
- Blood
 - Orange juice
 - Ammonia
 - Milk
- iv) Which substance has a pH value closest to neutral?
- Milk
 - Blood
 - Ammonia
 - Orange juice
- v) Which of the following would be most useful in testing if a substance is either an acid or a base?
- Filter paper
 - Litmus paper
 - Thermometer
 - Electronic balance